<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Partial Marks</th></tr>
<tr><td>5(a)</td><td>250</td><td>2</td><td>B1 for 5 [cm] oe</td></tr>
<tr><td>5(b)</td><td>Correct point E joined to A and D with ruled lines and with arcs</td><td>3</td><td>B2 for correct point E with arcs without lines

or correct ruled shape without arcs

or B1 for drawing AE = 11 cm or drawing DE = 12 cm

or correct point E without arcs and lines</td></tr>
<tr><td>5(c)(i)</td><td>Correct ruled bisector of angle ABC which reaches DE with two correct pairs of arcs</td><td>B2</td><td>B1 for a correct ruled angle bisector with no/wrong arcs or two correct pairs of arcs</td></tr>
<tr><td>5(c)(ii)</td><td>Correct ruled perpendicular bisector of side CD which reaches AE with two correct pairs of arcs</td><td>B2</td><td>B1 for a correct ruled perpendicular bisector with no/wrong arcs or two correct pairs of arcs</td></tr>
<tr><td>5(d)(i)</td><td>Constructed circle, centre 7 cm from B along bisector of ABC, with radius 3 cm</td><td>3</td><td>3FT along their (c)(i)
B1 for a circle, centre 7 cm from B, any radius
M1 for a circle, radius 3 cm seen anywhere</td></tr>
<tr><td>5(d)(ii)</td><td>942 or 943 or 942.4 to 942.6</td><td>2</td><td>M1 for $(2 \times 150)\pi$ or 300π soi</td></tr>
<tr><td>6(a)</td><td><table><tr><th></th><th>F</th><th>G</th><th>S</th><th>I</th><th>J</th><th>Tot</th></tr><tr><td>B</td><td></td><td>21</td><td></td><td></td><td>8</td><td></td></tr><tr><td>G</td><td>30</td><td></td><td></td><td>11</td><td></td><td>139</td></tr><tr><td>Tot</td><td>57</td><td></td><td>102</td><td></td><td>20</td><td></td></tr></table></td><td>3</td><td>B2 for 6 or 7 correct

or B1 for 3, 4 or 5 correct</td></tr>
<tr><td>6(b)(i)</td><td>$\frac{54}{\textit{their}139}$ oe isw</td><td>1</td><td>FT their table</td></tr>
<tr><td>6(b)(ii)</td><td>$\frac{46}{123}$ oe isw</td><td>1</td><td></td></tr>
<tr><td>6(b)(iii)</td><td>$\frac{209}{262}$ oe isw</td><td>1</td><td></td></tr>
<tr><td>6(c)(i)</td><td>[Chemistry] 80°
[Physics] 155°</td><td>2</td><td>B1 for each
or if 0 scored M1 for 125 ÷ 25 or 360 ÷ 72 or 5

If 0 scored SC1 for the two angles adding to 235°</td></tr>
</table>

Question	Answer	Marks	Partial Marks
6(c)(ii)	Two correct lines on the pie chart	**2**	**2FT** only if **(c)(i)** angles total 235° **B1** for a correct sector of 125° or 80° or 155°
7(a)	10 10	**1**	
7(b)(i)	22.4	**1**	
7(b)(ii)	6.2	**2**	**2FT** *their* **(b)(i)** × 1000 ÷ (60 × 60) oe rounded to 1dp or **M1** for *their* **(b)(i)** × 1000 ÷ (60 × 60) oe or 5600 ÷ (15 × 60) oe
7(c)	Two correct ruled lines	**2**	**B1FT** for a line (09 55, 0) to (*their* **7(a)**, 5.6) **B1FT** for horizontal line (*their* **7(a)**, 5.6) to (*their* **7(a)** + 23, 5.6)
7(d)(i)	12	**2**	**M1** for 5.6 ÷ 28 [× 60]
7(d)(ii)	Correct line	**1**	**FT** line from (10 07, 0) to (10 07 + *their* **(d)(i)**, *their* 5.6)
7(e)(i)	Correct line	**1**	**FT** line from (*their* **7(a)** + 23, 5.6) to (10 54, 0)
7(e)(ii)	16	**2**	**2FT** 5.6 ÷ (*their* time in minutes) × 60 **M1** for 5.6 ÷ 21 [× 60] soi or for 5.6 ÷ (*their* time in minutes)[× 60]
8(a)(i)	116	**1**	
8(a)(ii)	32	**1**	**FT** (180 – *their* **(a)(i)**) ÷ 2
8(b)(i)	Pentagon	**1**	
8(b)(ii)	Angle [between] tangent [and] radius	**1**	
8(b)(iii)	108 Angles [on a straight] line [add up to] 180	**2**	**B1** for angle **B1** for reason
8(b)(iv)	72	**1**	